

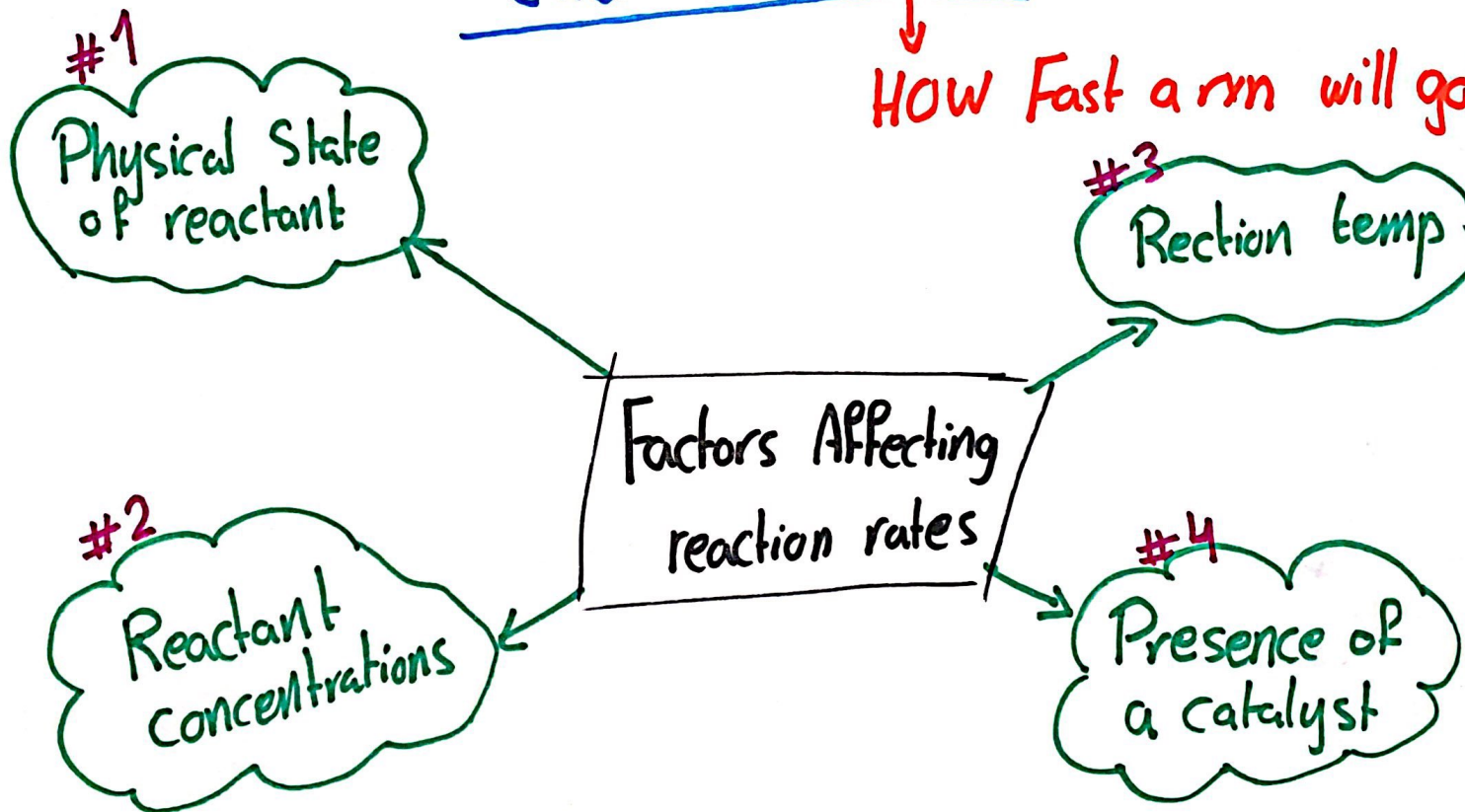
بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Ch.14: Chemical Kinetics

Part 1

Chemical Kinetics

↓
How Fast a rxn will go?



Rate: change in concentration with change in time

disappearance of reactant → appearance of products

$$\text{Rate} = \frac{\Delta []}{\Delta t} \text{ M/s}$$

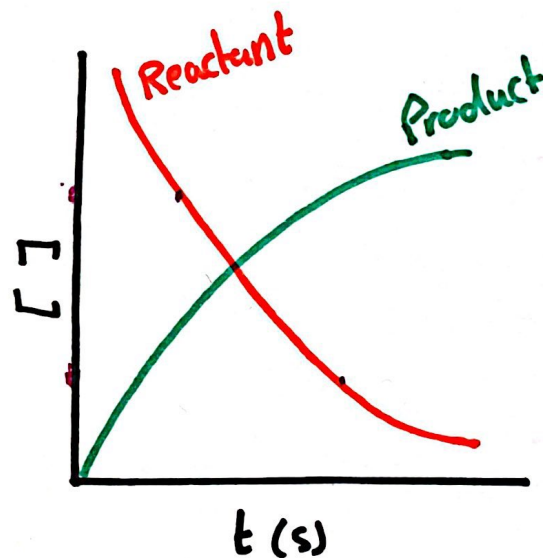


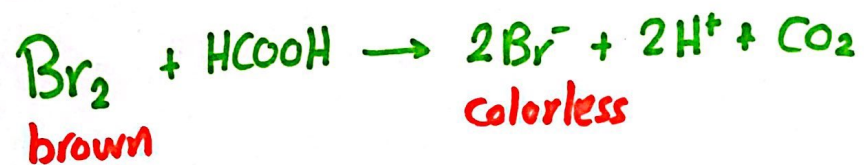
- concentration of reactants dec with time.

$$\text{rate} = -\frac{\Delta [A]}{\Delta t} = -\frac{[A]_f - [A]_i}{t_f - t_i}$$

- concentration of products inc with time.

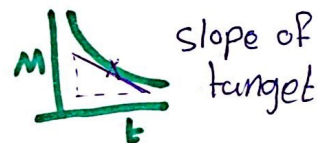
$$\text{rate} = \frac{\Delta [B]}{\Delta t} = \frac{[B]_f - [B]_i}{t_f - t_i}$$





- initial rate: rate at $t=0$

- instantaneous rate: rate at specific instance in time.



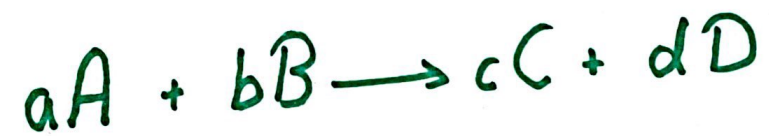
time (s)	$[\text{Br}_2]_M$
0	0.012 ← initial rate
50	0.0101
100	0.00846
150	0.0071
200	0.00596

- Average rate: rate over a specified period of time.

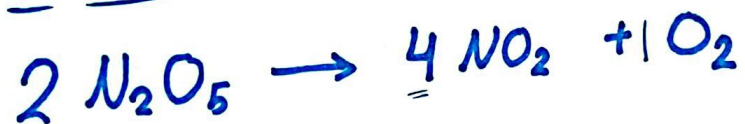
$$\text{average rate} = \frac{[]_f - []_i}{t_f - t_i}$$

Find the rate between 0s $\rightarrow$ 100s .

$$\begin{aligned}\text{rate} &= - \frac{\Delta[\text{Br}_2]}{\Delta t} \\ &= - \frac{0.00846 - 0.012}{100 - 0} \\ &= 3.54 \times 10^{-5} \text{ M/s}\end{aligned}$$



$$\begin{aligned} \text{rate} &= \frac{\text{rate } A}{a} = \frac{\text{rate } B}{b} = \frac{\text{rate } C}{c} = \frac{\text{rate } D}{d} \\ &= -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t} \end{aligned}$$



$$\begin{aligned} \text{rate} &= \frac{\text{rate } N_2O_5}{2} = \frac{\text{rate } NO_2}{4} = \frac{\text{rate } O_2}{1} \\ &= -\frac{1}{2} \frac{\Delta[N_2O_5]}{\Delta t} = \frac{1}{4} \frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t} \end{aligned}$$

Rate law

How the rate of the rxn changes with concentration of Reactants.



the order of the reaction with respect to A.

$$\text{rate} = k [A]^x [B]^y$$

the order of the rxn with respect to B

rate constant

depends on:

- specific rxn.
- temp.
- presence of catalyst.

$$\Rightarrow k \uparrow \text{ rate } \uparrow$$

$$\Rightarrow \text{unit: } M^{1-L} s^{-1}$$

L: overall order

$\frac{x+y}{\text{overall order of the reaction}}$

orders : determined

Experimentally

negative
positive
fractions

Determine the Rate law for this reaction:



$$\text{Rate} = 1.2 [\text{F}_2]^1 [\text{ClO}_2]^1$$

$$\text{rate} = k [\text{F}_2]^x [\text{ClO}_2]^y$$

$[\text{F}_2]$ M	$[\text{ClO}_2]$ M	initial rate (M/s)
0.1	0.01	1.2×10^{-3}
0.1	0.04	4.8×10^{-3}
0.2	0.01	2.4×10^{-3}

$$4.8 \times 10^{-3} = k [0.1] [0.04]$$

$$k = \frac{4.8 \times 10^{-3}}{0.1 \times 0.04} = 1.2 \text{ M}^{-1} \text{ s}^{-1}$$

$$x = 1$$

$$y = 1$$

$$L = 1 + 1 = 2$$

$$\text{unit of } k = \text{M}^{-L} \text{ s}^{-1}$$

$$= \text{M}^{-1} \text{ s}^{-1}$$

$$\frac{\text{rate 2}}{\text{rate 1}} = \frac{k \cancel{[0.1]^x} [0.04]^x}{k \cancel{[0.1]^x} [0.01]^x} = \frac{4.8 \times 10^{-3}}{1.2 \times 10^{-3}}$$

$$\left(\frac{0.04}{0.01}\right)^x = \frac{48}{12}$$

$$(4)^x = 4$$

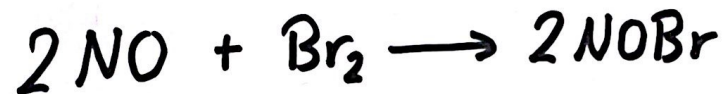
$$\boxed{x=1}$$

$$\frac{2.4 \times 10^{-3}}{1.2 \times 10^{-3}} = \frac{k(0.2)^x (0.01)^x}{k(0.1)^x (0.01)^x}$$

$$2 = \left(\frac{0.2}{0.1}\right)^x$$

$$2 = 2^x \Rightarrow \boxed{x=1}$$

Determine the Rate Law for this reaction:



$$\text{Rate} = k[\text{NO}]^x [\text{Br}_2]^y$$

[NO]	[Br ₂]	Rate Rate (M/s)
0.1	0.1	12
0.1	0.2	24
0.3	0.1	108

$x=1$
 $x=2$
 $L=3$
 unit $k = \text{M}^{-L} \text{S}^{-1}$
 $= \text{M}^{-2} \text{S}^{-1}$
 $= \text{mol}^{-2} \cdot \text{s}^{-1} / \text{L}^{-2}$
 $= \left(\frac{\text{mol}}{\text{L}} \right)^{-2} \text{S}^{-1}$
 $\text{mol}^{-2} \cdot \text{L}^2 \text{S}^{-1}$

$3^x = 9$
 $x=2$
 $\text{rate} = 12000 [\text{NO}]^2 [\text{Br}_2]^1$

$12 = k [0.1]^1 [0.1]^2$
 $k = \frac{12}{0.1^3}$
 $= 12000 \text{ M}^{-2} \text{S}^{-1}$

$$\frac{24}{12} = \frac{\cancel{K} [\cancel{0.1}]^x [0.2]^x}{\cancel{K} [\cancel{0.1}]^x [\cancel{0.1}]^y}$$

$$2 = \left(\frac{0.2}{0.1}\right)^x$$

$$2 = 2^x \Rightarrow \boxed{Y=1}$$

$$\frac{108}{12} = \frac{\cancel{K} [0.3]^x [\cancel{0.1}]^x}{\cancel{K} [0.1]^x [\cancel{0.1}]^1}$$

$$9 = \left(\frac{0.3}{0.1}\right)^x$$

$$9 = 3^x$$

$$\boxed{3^2 = 3^x} \quad \boxed{X=2}$$